

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Machine Learning Systems Aug 2025 to May 2026

- Automated provisioning, configuration, and **telemetry** for containerized test clusters, cutting experiment setup effort **55%** and making every sweep reproducible from config.
- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.
- Built an automated evaluation harness for controlled experiments, cutting setup effort **55%** and making every result reproducible from a single config.

Turito | Software Engineering Intern, Systems & Performance Jun 2022 to Oct 2022

- Slashed p95 CPU **22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Lowered network egress **14%** and p95 CPU **22%** on a high-frequency event path by removing redundant processing, payload copies, and per-packet logging overhead.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.

TECHNICAL SKILLS

Languages: C++17, Python, Rust, TypeScript, SQL

Retrieval & Indexing: HNSW written from scratch, AVX2/SIMD kernels, int8 quantization with exact re-ranking, ANN recall/QPS benchmarking, two-tower retrieval, embeddings, pybind11, Qdrant

Performance: CPU, memory and lock-contention profiling, cache-aware data layout, load testing, p50/p95 latency tuning

Systems: FastAPI, Docker, Kubernetes, AWS, Linux, CI/CD, Git, telemetry pipelines

PROJECTS

Proxima, Vector Search Engine | C++17, AVX2/SIMD, HNSW, pybind11

- Reached **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** hnswlib and **2.5x** FAISS, with an HNSW engine written from scratch in C++17.
- Cut traversal memory **75%** (51 MB to 12.8 MB) with **int8** AVX2 traversal plus exact float re-ranking and cache-aware node reordering.

ReelRank, Two-Stage Recommender | PyTorch, C++ HNSW, FastAPI, React, TypeScript

- Built a **two-tower** retrieval model with ANN candidate generation and a neural ranker, evaluated on a leakage-free temporal split with Recall, NDCG, and MAP.

Relay, LLM Inference Gateway | Python, FastAPI, asyncio, SQLite, SSE

- Cut p50 latency from **759 ms to 44 ms** over a **4,000**-request load test with an OpenAI-compatible gateway doing exact and **semantic caching**, failover, and rate limiting.

OPEN SOURCE

- microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)
- fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Distributed Computing, Computer Networks, Advanced Algorithms, Database Systems, Machine Learning, GPU Programming

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)